

Precept 1 Outline

ECO 202 TAs

September 10, 2026

1 Introductions (5 min)

- TA: name, year, where they are from, what their research is about, and answer a quick icebreaker (TA's choice)
- Students: their name, whether they have taken a stats class before, same icebreaker question as above

2 Overview & expectations for precept (5 min)

1. Purpose of precept:

- We will use precept to run through key practice problems and explain solutions on the board. Our goal will be to focus on statistical concepts themselves (as opposed to other goals of this course, like using AI tools ethically and effectively).
- We will post outlines like this document on GitHub: [Link to folder](#)

2. Tech policy:

- Please, no laptops or phones unless we are doing an activity together and ask you to try it on your computer, or something like this. The only materials you should typically need is something to take handwritten notes with (iPads are fine).

3. If you miss a precept:

- Ask a classmate for notes for reference. You can also try the problems listed on the posted precept document from that week, check what the solutions look like, and come to office hours with any remaining questions.

4. Asking questions has positive externalities!

- No question is a “dumb question.” We are happy to see you asking questions! Plus, it is helpful for you and for anyone else in the class to hear the question and the answer. So, please ask questions often and without hesitation.

5. We as TAs can't be 1:1 tutors, but there are multiple options for extra support:

- Come to office hours for any preceptor, as much as you want.
 - Dexin: Tuesdays, 4-5pm, tables outside of JRR A19
 - Emily: Thursdays, 4:30-5:30pm, JRR A74
 - Ryan: Fridays, 2-3pm, JRR A74
- Come to office hours of the professor: Fridays, 3pm-4pm, JRR 278
- Check out the McGraw Center:
 - Individual tutoring: [Link to page](#)
 - Group study with a tutor: [Link to page](#)

- Reach out to us by email if you've tried the above but are seeking support beyond these things.
- Email us if you have questions that can't be answered in precept, or in office hours.
 - If you email a TA, we can usually get back to you within 24-48 hours.
 - A response may take longer if you reach out late at night, or over the weekend.
 - Quick admin points to check in about:
 - Would you be able to attend the adjacent precept time in this block? (Your TA will explain – our sections are currently pretty small, so we are considering consolidating.)
 - If you are requesting an accommodation with ODS, please do that as soon as possible. We need to know about this for scheduling exam rooms, etc.
 - We are aware this course is experimental in how it is structured:
 - We will periodically check in about how things are going in the class, but please don't hesitate to ask us questions about the course itself or tell us your thoughts.
 - For example, in office hours we could talk about how to set up your workflow for taking notes & doing practice problems.
 - Any questions about the course before jumping in to this week's material?

3 Review of practice questions (40 min)

Main goal today: Practice with concepts related to describing a dataset and describing a distribution. We will also start practicing how to tell the difference between observational and causal evidence (but we will get into this in more detail after Lecture 5).

- (From Practice 1, Question 4)

A classmate asks you: “Does staying longer with the same employer raise a worker's hourly wage?”

The same historical `wage1` file used in lecture contains `tenure`, measured in years with the current employer, and `wage`, measured in 1976 dollars per hour.

- Suppose that you created a scatterplot with `tenure` on the x-axis and `wage1` on the y-axis, and that the points on the scatterplot slope upward. Write the strongest one-sentence conclusion supported by that fact alone.
- Name two alternative explanations for an observed positive relationship and one kind of design information that would be needed before making a causal claim.

- (From Practice 2, Question 1)

A city housing survey has one row for each responding renter household. Its columns include `monthly_rent` in dollars per month, `bedrooms`, `neighborhood`, `rent_burdened` coded 1 when rent exceeds 30 percent of household income and 0 otherwise, `response_mode` coded web or telephone, and `survey_weight`.

- State the observational unit.
- Classify each listed variable as categorical or quantitative. Explain why the values 0 and 1 do not make `rent_burdened` a quantitative variable.
- Explain why a difference in rent between the two `rent_burdened` groups would be descriptive rather than evidence that the coding itself caused rent to change.

- (From Practice 2, Question 2) The ordered hourly wages, measured in dollars per hour, for six workers are

4, 6, 7, 8, 9, 20

Use the median-of-halves convention for quartiles.

- (a) Before calculating, predict whether the mean will be below, equal to, or above the median. Explain your prediction from the distribution's shape.
- (b) Calculate the mean, median, Q_1 , Q_3 , IQR , range, sample variance, and sample standard deviation.
- (c) State the units of the mean, IQR , variance, and standard deviation.
- (d) Calculate the lower and upper $1.5IQR$ fences. Which observation, if any, is flagged? What does the flag establish, and what does it not establish?
- (e) Replace 20 by 50. Calculate the new mean, median, and IQR ; determine whether the sample standard deviation increases or decreases; and use the result to explain resistance.
- (f) Name two features of the original six observations that would be lost if someone reported only the mean and standard deviation.

4. (From Practice 2, Question 5)

Two clinics record the following patient waiting times, in minutes:

A: 4, 5, 6, 7, 8, 9, 18

B: 4, 5, 6, 7, 8, 9, 48

- (a) Calculate the mean, median, Q_1 , Q_3 , and IQR for each clinic using the median-of-halves convention.
- (b) Without calculating both standard deviations exactly, determine which clinic has the larger standard deviation and justify your answer.
- (c) Apply the $1.5IQR$ rule to each distribution. Explain why receiving the same flag does not make the two largest waiting times equally unusual in magnitude.
- (d) Write two sentences comparing the centers, spreads, and tails. Do not propose a cause for the difference.

Related Extra Practice

1. Observational studies suggest that countries with more generous unemployment benefits have higher rates of unemployment. If countries were to make unemployment benefits less generous, would this lead to lower unemployment rates? Discuss using relevant jargon.
2. Is the phone area code of a cell-phone owner a quantitative or qualitative variable? Explain. (The area code is a three digit number that is used to direct phone calls to a particular area of the U.S.)
3. Suppose a survey collects data on people's birth year, social security number and marital status. Which of these three variables are quantitative, and which are categorical? Explain.
4. You download a data file about student test scores, in which each row is a student's standardized test score for a particular subject (e.g., math, science, reading, history) in a particular school year. Students take more than one subject and we see their scores from more than one school year. What is the observational unit of the data file?
5. Name two graphs that can be used to display the distribution of a categorical variable.
6. Construct a data set for $n = 6$ individuals of a quantitative variable such that the mean is 10, the median is 4, the first quartile is 2 and the third quartile is 7.
7. Suppose a quantitative variable takes on the 6 values $\{1, 3, 3, 5, 7, 8\}$. Draw a boxplot of this distribution with numbers on the y-axis.
8. A quantitative variable x is observed for $n = 6$ individuals and takes on the values 2, 4, 5, 7, 8, 13.
 - Sketch the boxplot of this distribution. Make sure you put numbers on the y-axis.
 - Is 13 an outlier according to the 1.5*IQR* rule? Explain.
 - Find another distribution of $n = 6$ values with the same median and interquartile range as x , but with mean equal to 20.

9. The distribution of the variable x for $n = 50$ individuals is as follows

x	6	3	4	2
frequency	25	13	10	2

that is, $x = 6$ for 25 individuals, $x = 3$ for 13 individuals, etc.

- (a) Sketch a boxplot of this distribution, with numbers on the y-axis.
 - (b) Sketch a histogram of this distribution, with the first bin starting at 2, and bin size equal to 2. Indicate "frequencies" on the y-axis.
 - (c) Consider the variable $y = -2x + 7$, that is for each individual i with x-value equal to x_i , $y_i = -2x_i + 7$. What fraction of the y values is larger than zero? Explain.
10. A quantitative variable measured for $n = 40$ individuals takes on the values -2 for 10 individuals, the value 4 for 29 individuals, and the value 12 for the remaining individual.
 - Compute the mean and standard deviation of this distribution.
 - Is the value 12 an outlier of this distribution according the rule-of-thumb for outliers? Explain.
 11. The members of a basketball team are attempting 20 free throws each. Let x_i denote the number of successful attempts for player i . Suppose the mean of x , $\bar{x}$, equals 12.6 and the sample standard deviation, s_x , equals 4.2. Let y_i be the number of failed attempts of player i .
 - What are the mean and standard deviation of y_i ?
 - How would a scatter plot of the variables x and y look like? Be as specific as you can.
 12. Suppose a data set records the income of individuals. There are $n_1 = 153$ records on women, and the average income of women is \$36,000. There are also $n_2 = 210$ records on men, and the men's average income is \$38,000. What is the average income of all of the $n_1 + n_2$ individuals?